

University of Rwanda: MATH I

Continuous Assessment Test II / (40 marks)

Date: 12th / December / 2024

Time: 2 hours

Question I. a. Given an arithmetic progression $\{u_n\} = (2, 5, 8, 11, \dots)$. Find u_{30} and s_{30} . (/2 marks)

b. Determine if the series $\sum_{n=1}^{\infty} \frac{1}{n^6}$ converges or diverges. (/2 marks)

Question II. a) Verify that the set $\{(1, 1, 1)^T, (2, 1, -3)^T, (4, -5, 1)^T\}$ is an orthogonal set in $\mathbb{R}^3$? (/3 marks)

b) In view of a) find the corresponding orthonormal set /3 marks.

c) Is the set $S = \{(1, 1)^T, (1, -1)^T\}$ a basis for $\mathbb{R}^2$? (/4 marks)

Question III. Determine if the three vectors $\vec{a} = (1, 4, -7)$; $\vec{b} = (2, -1, 4)$ and $\vec{c} = (0, 9, 18)$ lie in the same plane or not? (/3 marks)

Question IV. Show that the sequence $(2 + \frac{1}{n})_{n \in \mathbb{N}}$ is monotone and bounded (/4 marks).

Question V. Find the centre, radius and interval of convergence of the following power series (/5 marks)

$$\sum_{k=0}^{+\infty} \frac{(x-5)^k}{k^2}$$

Question VI. Find the characteristic polynomial, eigenvalues and their corresponding eigenvectors of the matrix (/5 marks)

$$M = \begin{bmatrix} 4 & -5 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix}$$

Question VII. Find the sum of the series (/4 marks)

$$\sum_{k=1}^{+\infty} \left[\left(\frac{2}{3} \right)^k + \frac{1}{k(k+1)} \right]$$

Question VII. Solve the following system of equations using matrix inverse method (/5 marks).

$$\begin{cases} x + 2y + 3z = 3 \\ 2x + 4y + 5z = 4 \\ 3x + 5y + 6z = 8 \end{cases}$$

GOOD LUCK